Contribution Journal

Module 10 Lab: Unsupervised Learning
Group: FA26_ML_6171_16347 5 | Course: ITAI 1371

Group Members and Assigned Sections

Team Member	Section	Notebook Part
Jamie Lopez	Section 1	Part 1: Supervised vs. Unsupervised Learning, data setup and exploration
Kerry Walker	Section 2	Task 1: Elbow Method for the optimal number of clusters
Angel Martinez	Section 3	Task 2: K-Means clustering and visualization
Kristina Akpan	Section 4	Task 3: PCA on the Iris dataset
Thomas Jaymon	Section 5	Extension: PCA + K-Means combined workflow, Knowledge Check, and final notebook review

Jamie Lopez

Section 1: Part 1, Supervised vs. Unsupervised Learning, data setup and exploration

My part was Section 1, and the first thing I had to do was fix the starter notebook, because its code cells had lost their line breaks and would not run. I rebuilt them line by line and put all the shared imports (numpy, pandas, matplotlib, seaborn, and make_blobs) at the top so everyone else's sections would run without errors. Then I generated the synthetic customer dataset of 300 samples with 4 centers, plotted it, and added a data exploration cell that prints the shape of X and a describe() summary. I wrote code comments explaining why the true labels in y are set aside and only used later for checking, and I wrote the Section 1 interpretation explaining that the two features are on a similar scale and that the model never receives the answer key.

Kerry Walker

Section 2: Task 1, Elbow Method for the optimal number of clusters

Section 2: Elbow Method. Code: a loop fitting K-Means for k = 1 through 10, storing each inertia value, printing the values in a table, and plotting the elbow curve with labeled axes and a tick at every k. Added: a silhouette score analysis for k = 2 through 10 (slide 9) to confirm the choice of k; best result k = 4 with a score of 0.834. Written: the Section 2 interpretation, covering the sharp inertia drop through k = 4 (362.47), the flattening afterward, and the agreement between the elbow and silhouette methods.

Angel Martinez

Section 3: Task 2, K-Means clustering and visualization

I handled Section 3, which is where the actual customer segments get built. I fit the final K-Means model with k = 4, pulled out the cluster labels and centroids, and made the cluster scatter plot with

the centroids marked as red X markers. I added a summary cell that shows each centroid's coordinates, how many customers landed in each cluster (75 each), how many iterations it took to converge (3), and the final inertia. I also checked our clusters against the hidden true groups with the Adjusted Rand Index (1.000) and a crosstab, and I wrote the Section 3 interpretation explaining why cluster numbers are just names and why the Adjusted Rand Index was a better choice than plain accuracy.

Kristina Akpan

Section 4: Task 3, PCA on the Iris dataset

I completed Section 4, PCA on the Iris dataset. I reduced the 4 Iris features to 2 principal components, printed the explained variance (92.46% and 5.31%, for a total of 97.77%), and created the PCA scatter plot colored by species. I fixed a missing seaborn import in the starter code and changed the legend to show species names instead of numbers. I added a cell that prints the component loadings and the cumulative variance for 1 to 4 components, and another cell comparing PCA on standardized features (95.81% total) to show the scaling effect from slide 24. I wrote the Section 4 interpretation covering what PC1 represents, how the species separate, and why scaling changes the results.

Thomas Jaymon

Section 5: Extension combining PCA and K-Means, Knowledge Check, and final notebook review

My contribution consisted of Section 5. I designed and implemented an extension applying the combined workflow described on slide 25: K-Means with k = 3 was performed on the two principal components produced in Section 4, and the resulting clusters were plotted alongside the true species for comparison. I evaluated the agreement using the Adjusted Rand Index (0.716), compared it with K-Means on all four original features (0.730), and produced a species by cluster crosstab demonstrating that setosa was clustered perfectly while 14 virginica specimens were grouped with versicolor. I authored the interpretation of the extension and all three Knowledge Check responses, drawing on the inertia values from Section 2, the 97.77 percent explained variance from Section 4, and the results of my extension. Finally, I reviewed the complete notebook to confirm that each interpretation corresponded to its output and that every attribution label matched the Group Members table.